



SOFTWARE DEVELOPMENT WITH JAVA (SESSIONAL) — CSE-202

Project Lighthouse

Wellbeing and Self-Reflection Web Platform

COURSE

Software Development with JAVA (Sessional)
(CSE-202)

DEPARTMENT

Computer Science & Engineering, CUET

DEVELOPMENT TEAM

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Why Does Mental Wellbeing Go Untracked?



No Structured Habit

People want to reflect on their mood and mental state daily, but have no guided framework to do it consistently.



Fear of Privacy Invasion

Most organisational wellness tools are rejected by users because they fear their personal reflections are being read by managers.



No Longitudinal Insight

Standalone mood apps capture a single moment but never correlate sleep, stress, activity, and behaviour over time.



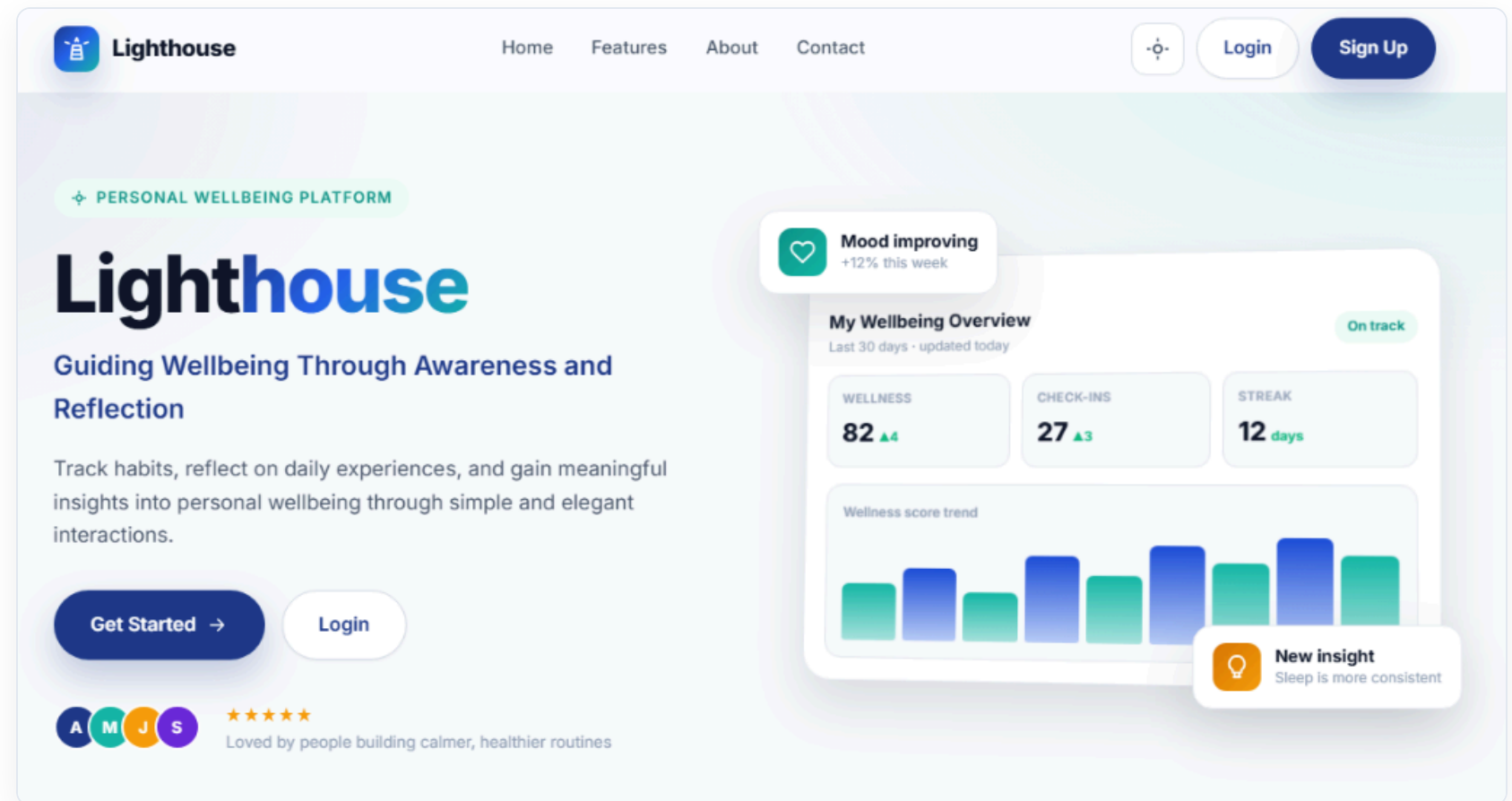
No Gentle Intervention

Admins or counsellors have no visibility into who may need a gentle check-in — until it's already too late.

INTRODUCTION

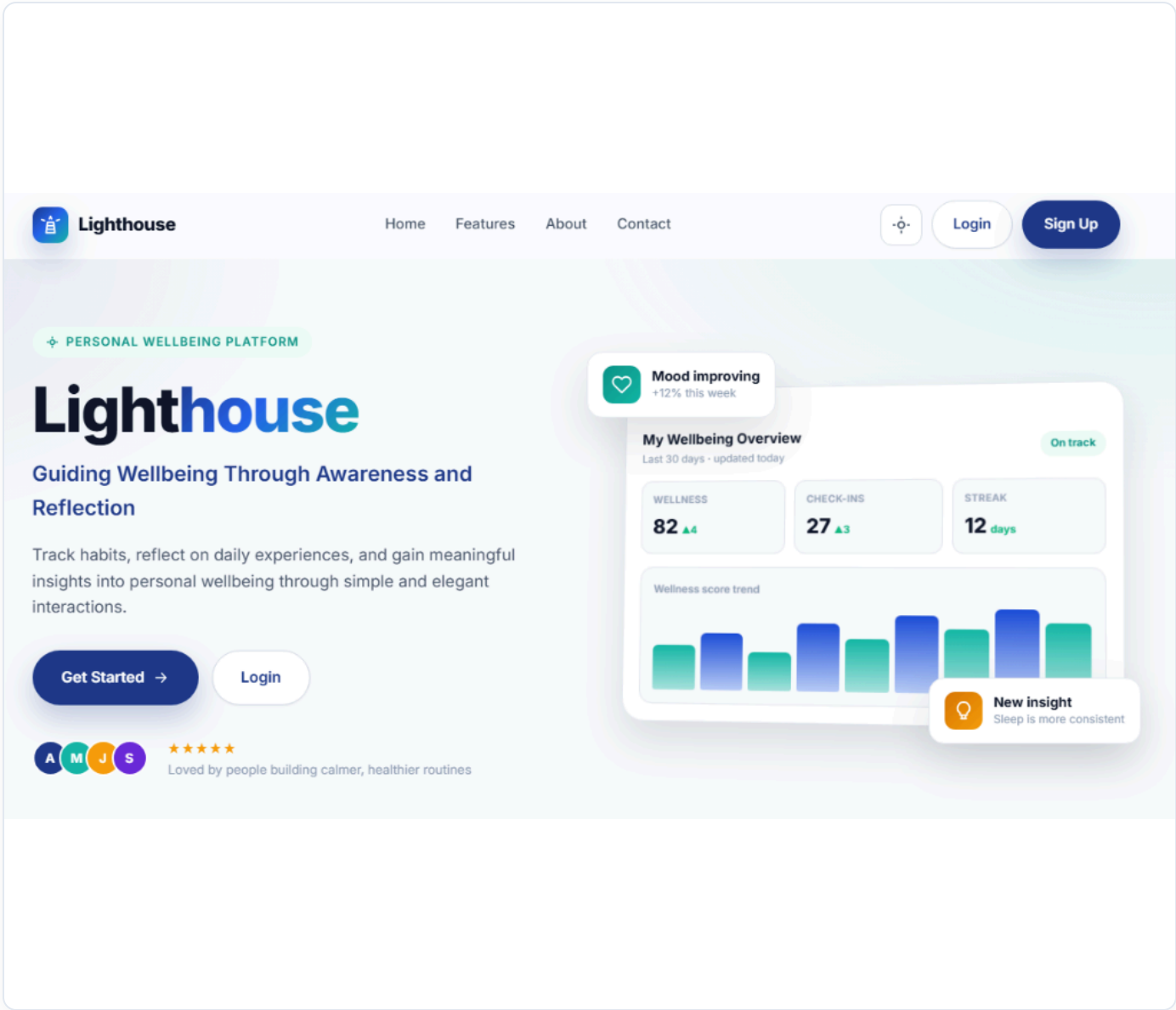
What is Lighthouse?

- A **personal wellbeing web app** that guides users through a short daily journey of check-in, scenario, visual reflection, and journaling.
- Tracks patterns over time and surfaces personal insights — mood, sleep, energy, productivity, behaviour.
- Includes a **separate Admin Portal** where staff can monitor overall platform health and support at-risk users — *without ever reading private journal entries*.
- Built for schools, organisations, or any community that needs to support people's mental wellbeing responsibly.



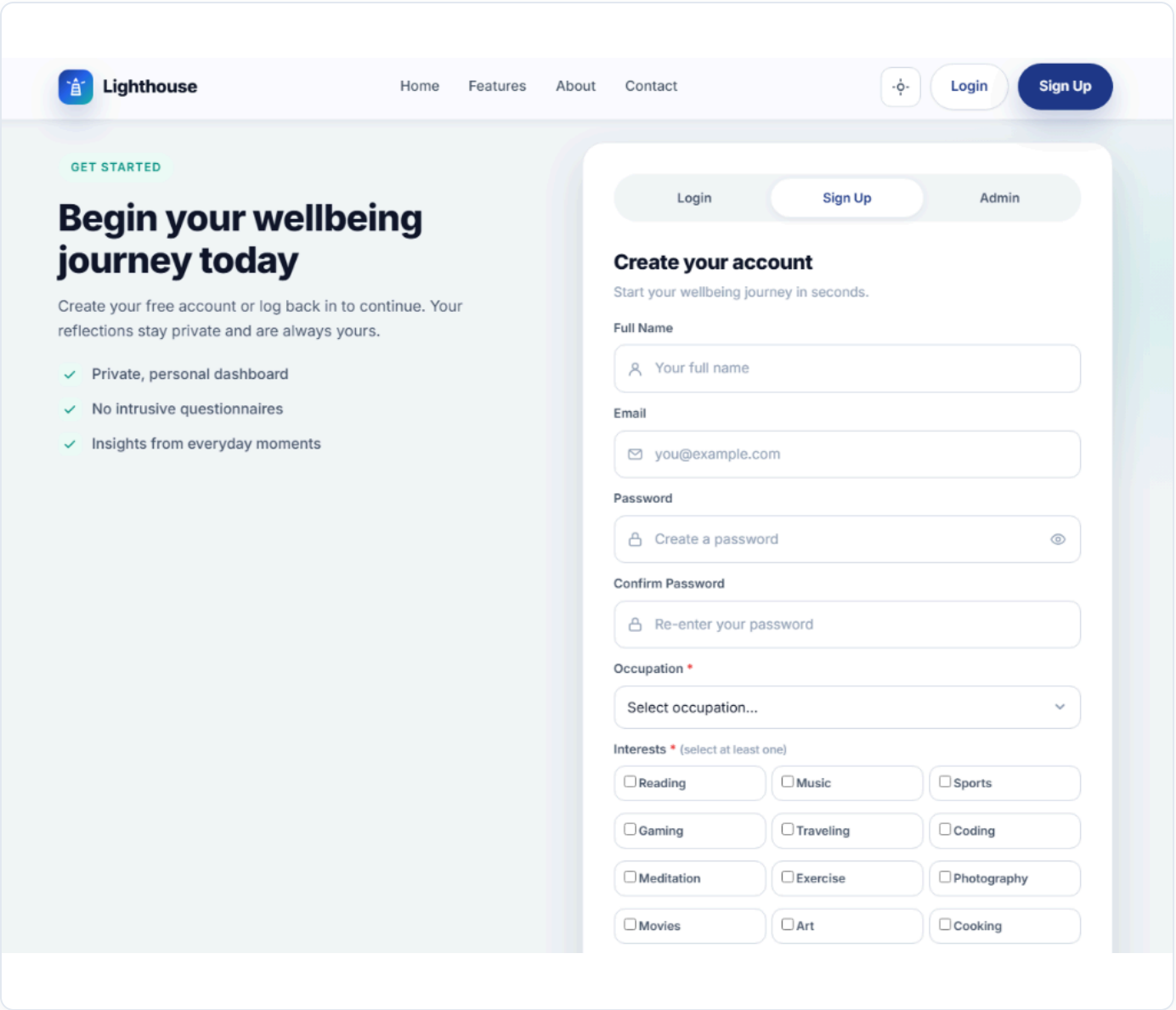
Lighthouse landing page — marketing + auth entry point

Landing Page & User Sign-Up



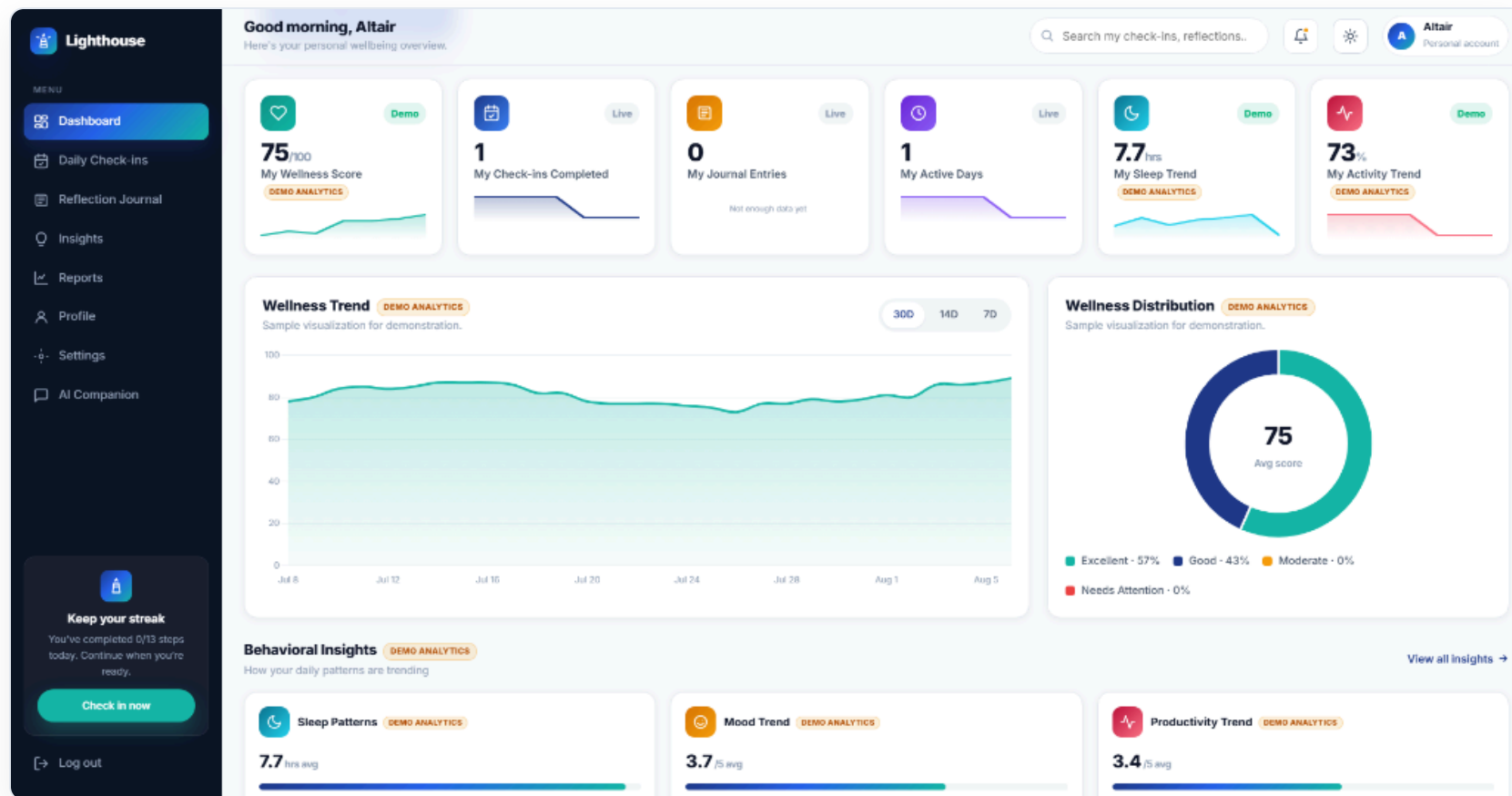
Marketing + Auth Page

A single page hosting Login, Sign Up, and Admin tabs. New visitors see the product pitch; returning users jump straight into the auth card.



Extended Sign-Up Profile

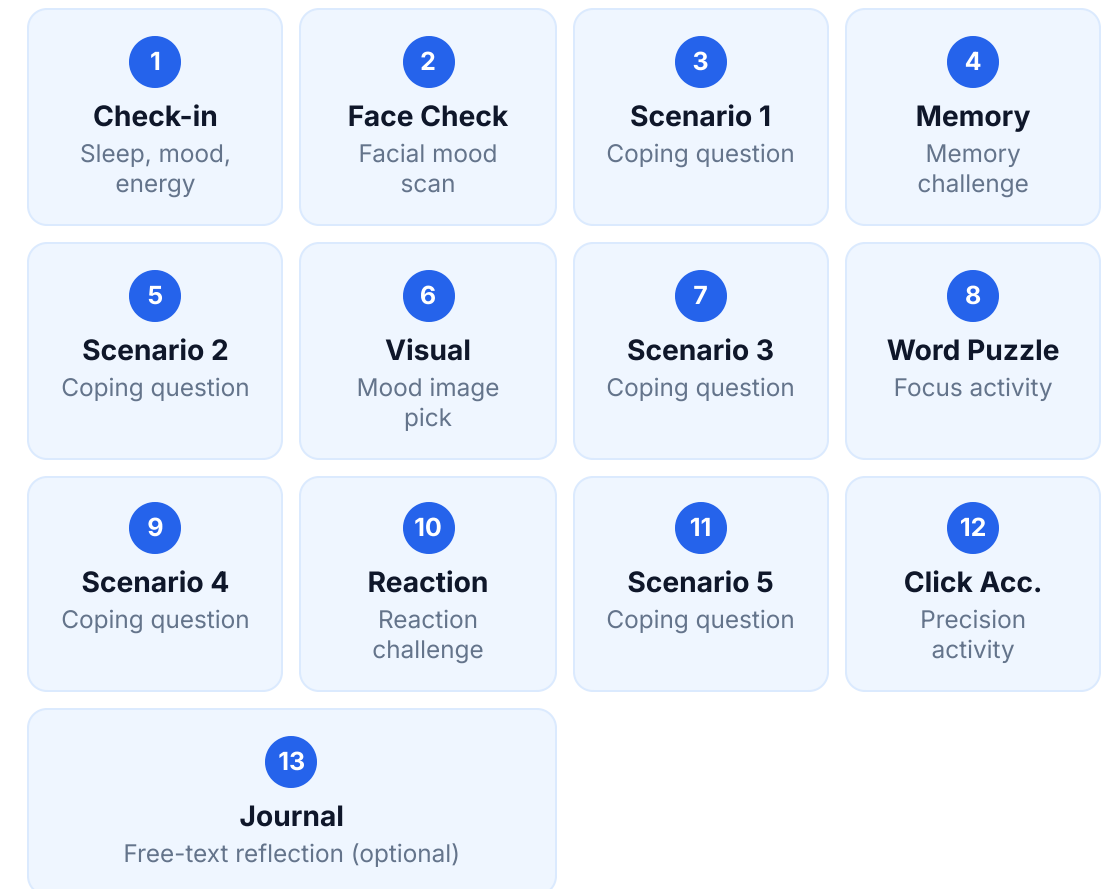
Sign-up collects extended profile fields — display name, avatar, occupation, interests, DOB — giving the AI Companion richer personalisation context from day one.



Dashboard — today's progress, live stats & analytics toggle

CORE LOOP

The Daily Journey — 13 Steps



- Steps unlock in sequence — each completed step reveals the next. Face Check & Journal are **optional**.
- Dashboard shows **Today's Progress** and links directly to the next incomplete step.

Daily Check-in & Scenario Assessment

Lighthouse

MENU

Dashboard

Daily Check-ins

Reflection Journal

Insights

Reports

Profile

Settings

AI Companion

Daily Check-ins

Quickly record today's wellbeing in one place.

Altair
Personal account

Today's wellbeing journey

Step 1 of 11 — after saving, you'll continue to the next journey step automatically.

Today's Check-in

Record how today feels so far.

Saved today

Sleep Hours (0-12)

e.g. 7.5

Mood

Select mood

Energy Level 3

Productivity 3

Physical Activity

Select level

Water Intake (optional)

e.g. 6 glasses

Today's Notes (optional)

Anything worth noting about today...

Delete today's entry

Save Check-in

Check-in History

Your previous daily check-ins, newest first

2026-08-05 · 😊 Good · Sleep 7h

Energy 3/5 · Productivity 3/5 · Activity Light · Water 5

Log out

Step 1 — Daily Check-in

Users log sleep hours, mood (1–5), energy, productivity, and physical activity. Structured sliders take under a minute and feed all longitudinal charts.

Scenario Assessment

Scenario 1 of 5 — sequential daily journey

Altair
Personal account

Guided journey

Five everyday situations today — complete them in order with short activities between. Not a diagnosis.

ACADEMIC PRESSURE · SCENARIO 1 OF 5

Exam Week Fog

Exam week is close and your notes feel messy. You are unsure where to begin studying.

What would you most likely do?

A

Create a simple priority list for the toughest topics first.

B

Reread everything from the beginning.

C

Study only what friends are studying.

D

Avoid studying until you feel ready.

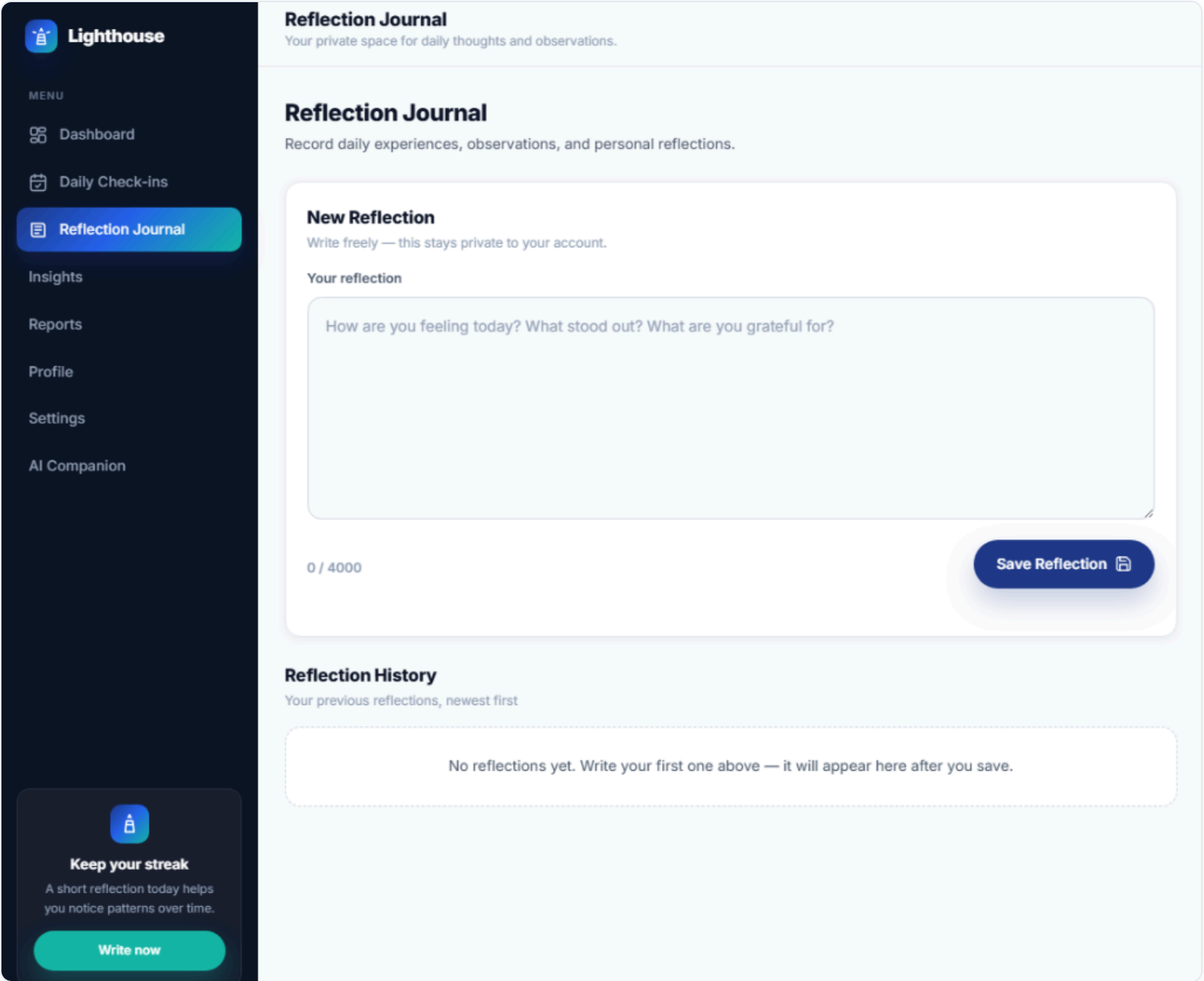
Steps 3, 5, 7, 9, 11 — Scenario Assessment (×5)

Five scenario rounds are woven throughout the journey, each presenting a short situational story. Drawn from an 80+ item bank, they probe behavioural coping patterns without clinical framing.

GUIDED JOURNEY — STEPS 6 & 13

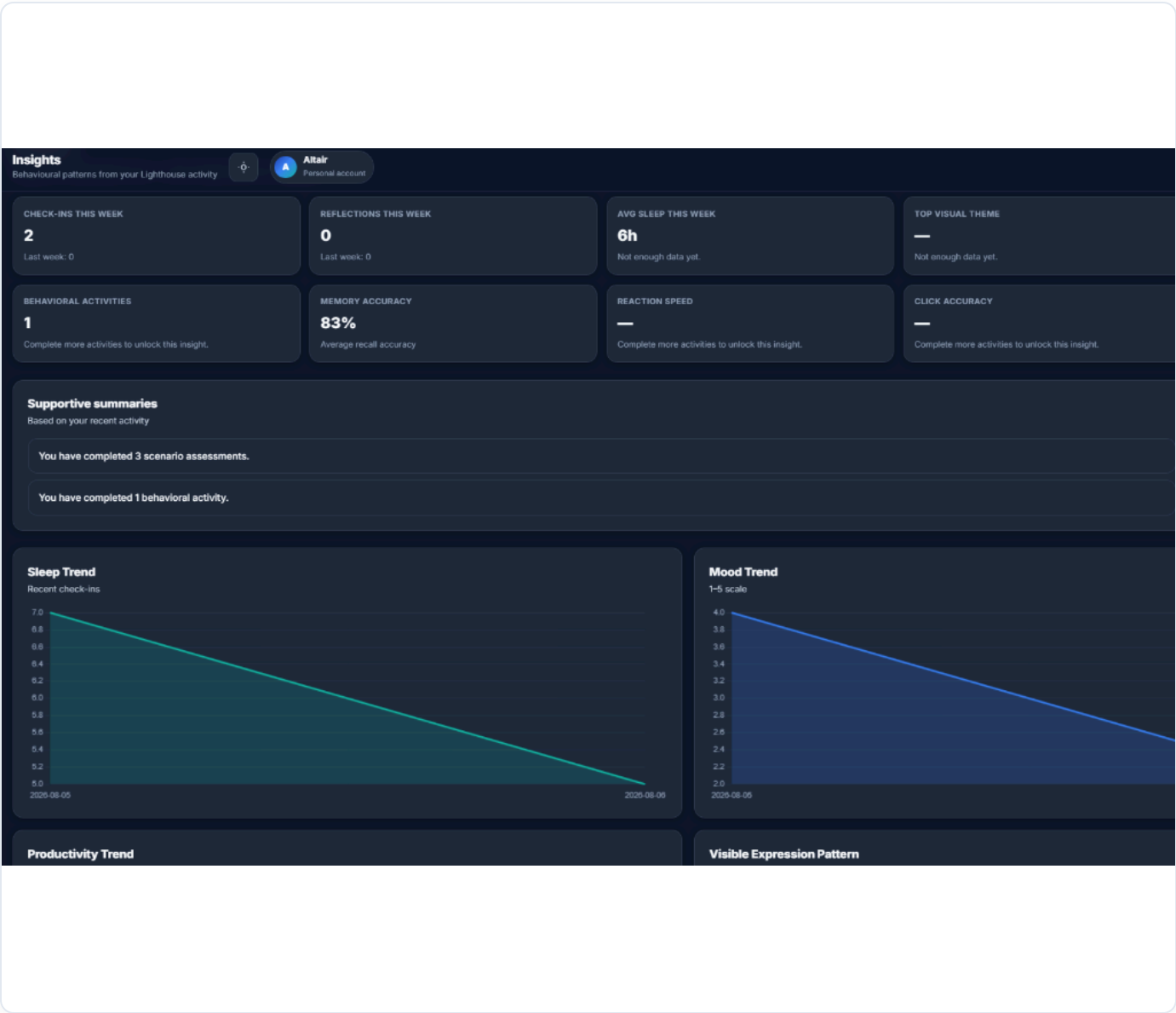
Visual Reflection & Private Journal

- **Step 6 — Visual Reflection:** User picks an image that matches their current emotional state — Calm, Heavy, Overwhelmed, Hopeful, etc. Adds a qualitative mood signal beyond numbers.
- **Step 13 — Reflection Journal (optional):** A free-text canvas for unconstrained personal writing. Guided prompt cards help users who aren't sure where to start.
- Journal text is protected by Supabase **Row Level Security (RLS)** — only the author's session can read it. Admins see no journal content, by design.



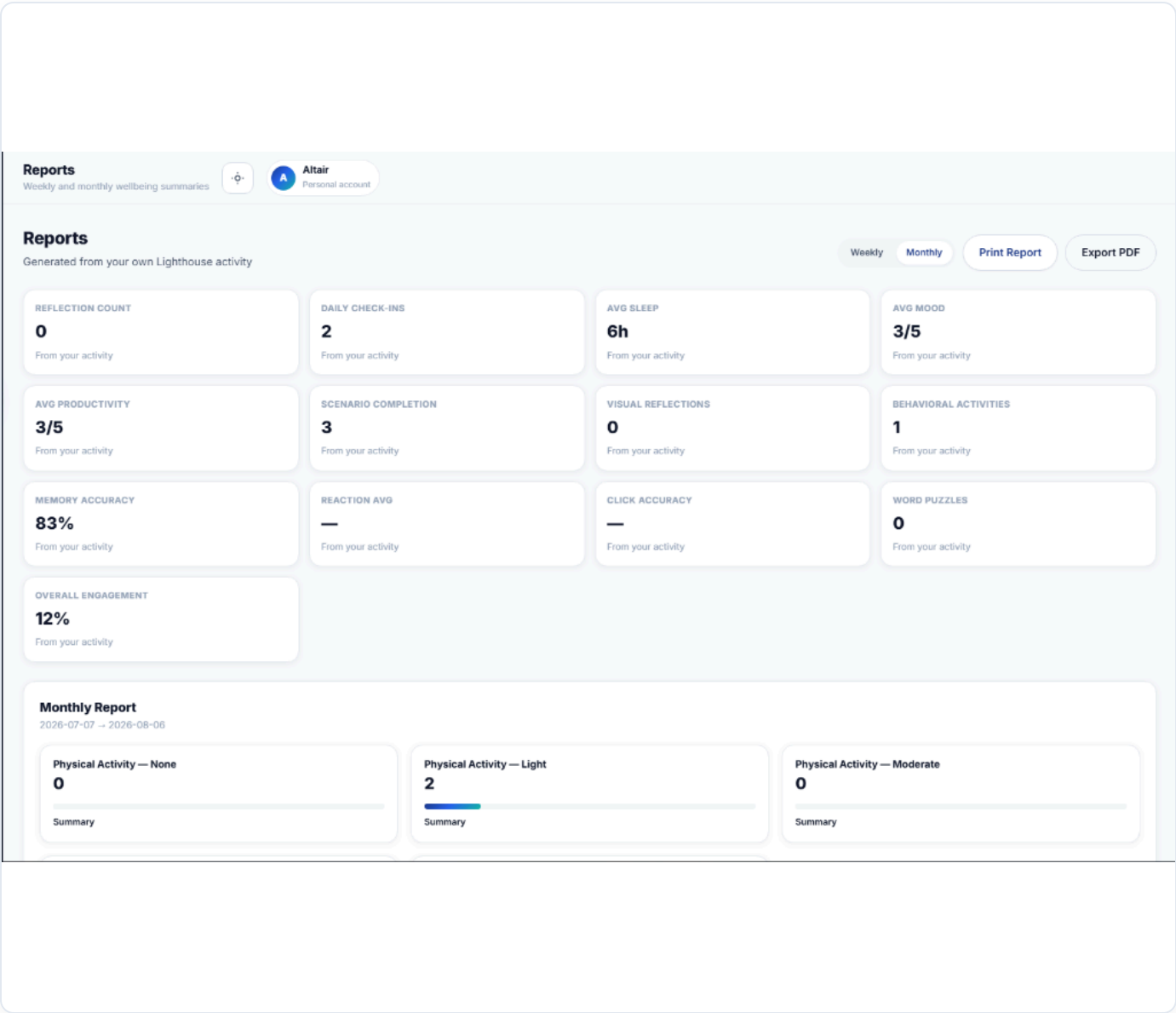
Reflection Journal — private, prompt-guided writing canvas

Insights & PDF Reports



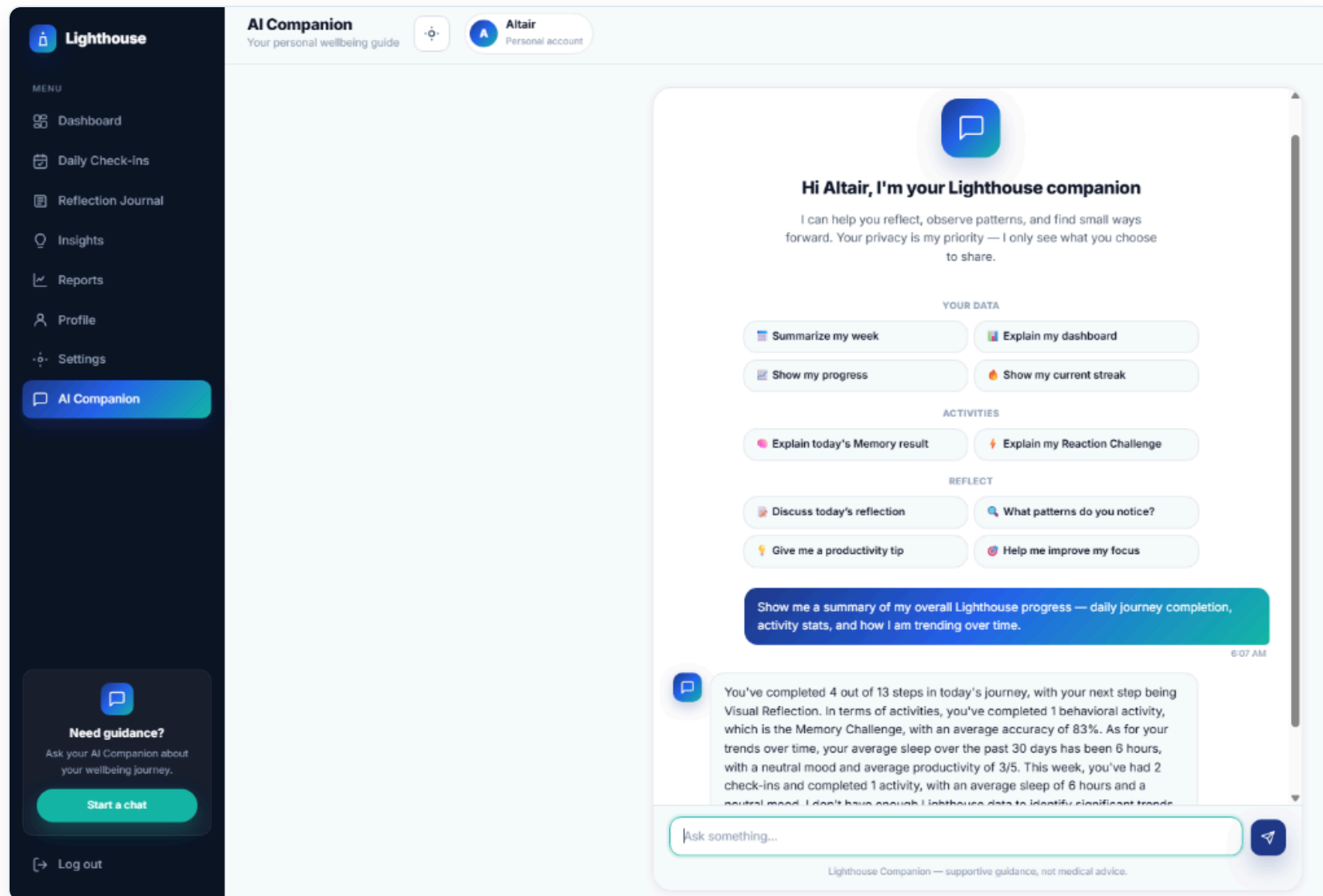
Insights Page

Behavioural pattern dashboard powered by Chart.js. Correlates sleep, mood, energy, productivity, and scenario scores across days and weeks — unlocking after 7+ check-ins.



Weekly / Monthly Reports + PDF Export

Summarises the user's journey into readable weekly and monthly snapshots. One-click PDF export (jsPDF) compiles all charts and scores for personal records or therapy consultations.



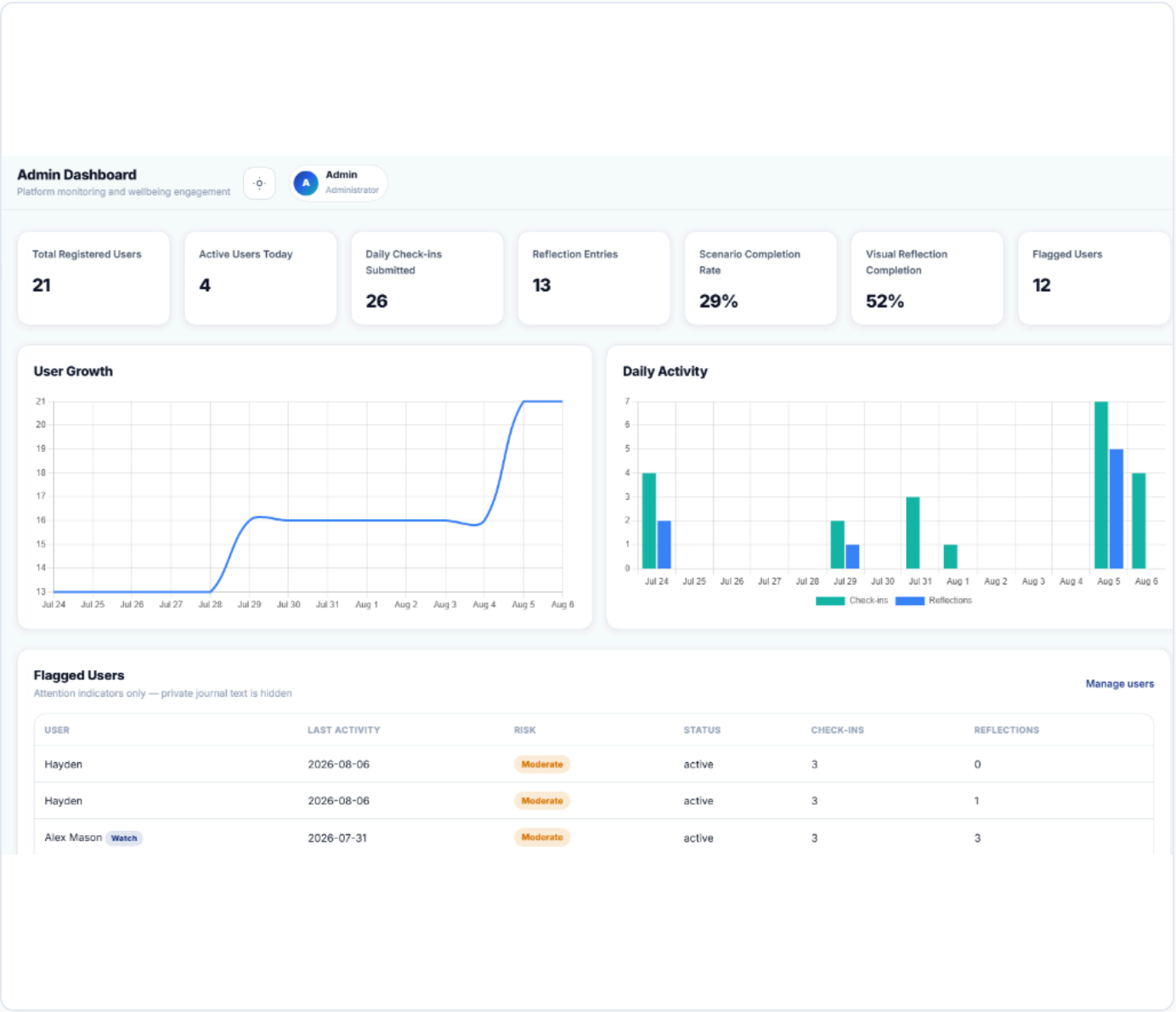
AI Companion — context-aware wellbeing chat

AI FEATURE

AI Companion — Lighthouse Chat

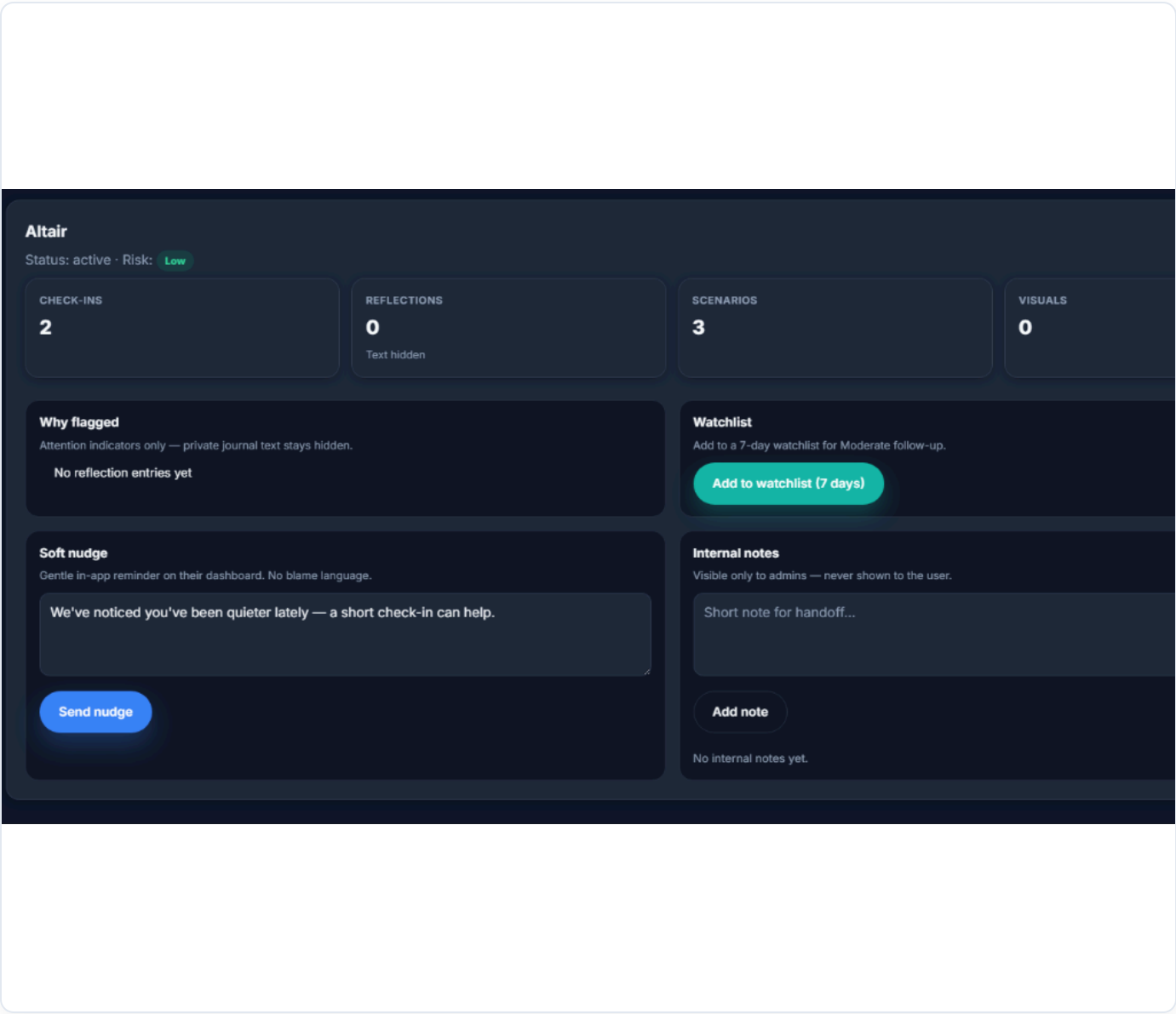
- Powered by **Groq's Llama-3.3-70B** model via a secure Express proxy server — the API key never reaches the browser.
- Receives a **live context snapshot** of the user's own data: dashboard stats, recent check-ins, journey progress, and streaks.
- Answers questions like "Summarise my week", "What's my mood trend?", or "What should I focus on today?" — *grounded in actual data, never fabricated*.
- Strictly scoped: cannot diagnose conditions, cannot see any other user's data, and refuses to invent scores.

Platform Monitoring & Care Panel



Admin Dashboard & Analytics

A bird's-eye view of the whole platform: total users, daily activity, growth charts, flagged counts, and PDF report packs. Journal text is never shown here by design.



Care Panel — Watchlist & Nudges

Flagged users (via heuristics: low mood streaks, inactivity, avoidance patterns) appear here. Admins review attention signals, add private notes, send a soft nudge, or place users on a 7-day watchlist.

User Directory & Scenario Management

User Management

Search and review account summaries (journal text hidden)

Admin

Administrator

Search by name or id...

All risk

NAME	USER ID	REGISTERED	LAST LOGIN	STATUS	RISK	
dfsdfsdfds	fb1c86e2...	Aug 5, 2026	Aug 5, 2026	active	Low	Care
Edward	664c8b63...	Aug 5, 2026	Aug 6, 2026	active	Low	Care
Altair	07145102...	Aug 5, 2026	Aug 6, 2026	active	Low	Care
David Mason	eb04c92e...	Aug 5, 2026	Aug 5, 2026	active	Low	Care
User	9733a140...	Aug 5, 2026	Aug 5, 2026	active	Low	Care
Hayden	8bec0b13...	Jul 29, 2026	Aug 6, 2026	active	Moderate	Care
Golam	27ed1882...	Jul 29, 2026	Aug 5, 2026	active	Low	Care
Mohammed Irfan	3b2870b2...	Jul 29, 2026	Jul 29, 2026	active	Moderate	Care
Hyre	0e1d6aa8...	Jul 24, 2026	Jul 24, 2026	active	Moderate	Care
Phillips	04c1ee07...	Jul 24, 2026	Aug 5, 2026	active	Low	Care

User Directory

Admins browse all registered users, view engagement status, and surface accounts that may need a care action — with no access to private journal or reflection content.

Lighthouse Admin

Scenario Management

Create, edit, enable or disable scenarios

Admin

Administrator

ADMIN

Dashboard

Users

Scenarios

Reports

Analytics

Settings

Edit Scenario

Title

Assignment Crunch

Category

Academic Pressure

Story

A university student has several assignments due tomorrow but has spent the evening scrolling through social media.

Question

What would you most likely do?

Option A

Break the work into smaller tasks and start with one

Option B

Continue scrolling and hope it works out.

Option C

Ask a classmate or friend for help organizing priorit

Option D

Leave everything until tomorrow morning.

☒ Enabled for users

Save Scenario

#

TITLE

CATEGORY

1

Assignment Crunch

Academic Pressure

Log out

Scenario Bank Management

Admins create, edit, enable, or disable scenario items. The bank ships with 80+ seed scenarios and can be expanded to match the organisation's context or user population.

Technology Stack



FRONTEND

Vanilla HTML / CSS / JS

Static multi-page app — intentionally no framework. Fast, zero-build, deployable to any static host.



DATABASE & AUTH

Supabase (Postgres + RLS)

Row Level Security enforces the privacy boundary at the DB layer — users can only see their own rows.



AI BACKEND

Groq · Llama-3.3-70B

Express.js proxy server holds the Groq API key server-side, feeds the model a live user data snapshot per request.



DATA VISUALISATION

Chart.js

Renders all mood, sleep, energy, productivity and behavioural trend charts on the dashboard and insights pages.



PDF EXPORT

jsPDF (client-side)

Generates wellness reports and admin PDF packs (Platform Snapshot, Engagement Pack, Flagged Care Summary) entirely in the browser.



HOSTING

Vercel (Static Deploy)

Deployed via vercel.json. Any static host works — no build step. Supabase config injected via config.js.

What Lighthouse Delivers

- **Guided Daily Habit:** A 13-step journey building consistent self-reflection — check-ins, scenarios, behavioural activities, visual, and journal.
- **Personal Insights:** Longitudinal charts surface mood-sleep-energy correlations that a single-point wellness check could never reveal.
- **AI Companion:** Context-aware Groq-powered assistant grounded in the user's real data — no hallucinated scores, no generic advice.
- **Bulletproof Privacy:** Supabase RLS means journal entries are cryptographically scoped to the user. Admins see signals, never diaries.
- **Compassionate Admin Care:** Heuristic flagging + soft nudges let admins reach out proactively — as a care tool, not a surveillance tool.
- **Future Roadmap:** Component framework migration, mobile app, and predictive anomaly detection for admin care signals.



Thank You

Questions & Discussion

PROJECT LIGHTHOUSE

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